

Proofs for “VioExplain: Single-Event Knowledge Explains Concurrent Industrial Constraint Violations”

This document proves the results stated in Section V of the main text. Each result is restated with the number it carries there, and the proofs use only the assumptions written in the statements. Proposition S1 on functional constraints, which Section IV-A of the main text cites, is stated and proved in Section 6.

1 Notation and the Path of Algorithm 1

A window X has the description $\phi = \phi(X) \in \mathbb{R}^D$ and the violation features $\mathcal{V}(X)$. The candidate events form the set $\mathcal{E}$ with $|\mathcal{E}| = K$. For an event E , $R(E) = R^*(E) \cup R^+(E)$ collects its exact and possible representations, and $R(E)$ also denotes the set of units of these representations. Under a common noise stream ξ , $X^{(0)}(\xi)$ is the fault-free run and $X^{(H)}(\xi)$ the run in which exactly the events of H act. The main text writes X_H and $X_\emptyset$ for windows of these runs.

Algorithm 1 of the main text visits the following path. Put $S_0 = \emptyset$ and $r_0 = \phi(X)$. If $p_\emptyset(X) \geq \tau_0$, the output is $\emptyset$. Otherwise $E_1 = \arg \max_{E \in \mathcal{E}} p_E(X)$, $S_1 = \{E_1\}$ and $r_1 = r_0 - \hat{\Delta}_{E_1}(r_0)$. For $j \geq 1$, write $z_E(r)$ for the addition score $z_E(X, r, S_j)$, whose window and selected set are fixed at step j . The candidate is $E_{j+1} = \arg \max_{E \notin S_j} z_E(r_j)$. If $z_{E_{j+1}}(r_j) \leq \tau_1$ or $|S_j| = K_{\max}$, the algorithm stops and outputs S_j . Otherwise $S_{j+1} = S_j \cup \{E_{j+1}\}$ and $r_{j+1} = r_j - \hat{\Delta}_{E_{j+1}}(r_j)$. The assignment step sends a violation on unit u to $\arg \max\{f_E(u) : E \in S, u \in \text{Supp}(E)\}$ and to the background E_0 when no selected event has u in its support.

2 Theorem 1 and Corollary 1

2.1 Model

The units form the set U , and unit u has the continuous degree $x_u \in \mathbb{R}$, which is defined below the violation threshold as well.

- (M1) Events are present independently with $\Pr(E \in H) = \pi_E \leq \frac{1}{2}$.
- (M2) Given H , a latent assignment $a : U \rightarrow H \cup \{\perp\}$ satisfies $a(u) = E \Rightarrow u \in R(E)$. The background $\perp$ may take any unit, except that $a(u) \neq \perp$ whenever $u \in R^*(E)$ for some $E \in H$, so a unit of an exact representation of a selected event is explained by some selected event, not necessarily by E .
- (M3) Given (H, a) , the degrees are conditionally independent with $x_u \sim p_u(\cdot \mid a(u))$, and $p_u(\cdot \mid \perp) = p_u^0$ is the normal density.

Write $g_u(E) = \log \frac{p_u(x_u \mid E)}{p_u^0(x_u)}$, $g_u(\perp) = 0$ and $w_E = \log \frac{1 - \pi_E}{\pi_E} \geq 0$. For a possible representation that E produces with probability $\Pr_E(u)$ and that otherwise follows the normal density, $p_u(x \mid E) = \Pr_E(u) p_u^{\text{resp}}(x) + (1 - \Pr_E(u)) p_u^0(x)$, so $g_u(E) \geq \log \Pr_E(u) + \log \frac{p_u^{\text{resp}}(x_u)}{p_u^0(x_u)}$. Using the right side in place of $g_u(E)$ can only underestimate the gain of a possible representation.

For $H \subseteq \mathcal{E}$ put $R^*(H) = \bigcup_{E \in H} R^*(E)$, $m_u(H) = \max\{g_u(E) : E \in H, u \in R(E)\}$ with $\max \emptyset = -\infty$, and

$$v_u(H) = \begin{cases} m_u(H), & u \in R^*(H), \\ \max(0, m_u(H)), & u \notin R^*(H), \end{cases} \quad F_{\text{ex}}(H) = \sum_{u \in U} v_u(H) - \sum_{E \in H} w_E.$$

The separable objective of the main text is

$$F(H) = \sum_{u \in U} \max(0, m_u(H)) - \sum_{E \in H} \hat{w}_E, \quad \hat{w}_E = w_E + \sum_{u \in R^*(E)} [-g_u(E)]_+.$$

Theorem 1 (Maximum a posteriori explanation). *Assume (M1) to (M3). Then for every H the separable objective $F(H)$ is a lower bound of the largest log posterior ratio of (H, a) over assignments a against the background explanation, with equality unless a unit of an exact representation has a negative log-likelihood ratio under a selected event that owns it and is covered by another selected event, and F is a monotone submodular function minus a modular one. With $G_u = \max_{E: u \in R(E)} [g_u(E)]_+$, the quantity $\sum_u G_u - F(H)$ is the cost of an uncapacitated facility location problem with assignment costs $G_u - [g_u(E)]_+$, opening costs $\hat{w}_E$ and a background facility with zero opening cost. Maximizing F is NP-hard, its cost admits no $(1 - o(1)) \ln|U|$ approximation unless $P = NP$, and with binary responses the model reduces to parsimonious covering.*

Proof. Step 1, the exact posterior. By (M1) to (M3),

$$\log \Pr(H, a, \phi) = \sum_{E \in H} \log \pi_E + \sum_{E \notin H} \log(1 - \pi_E) + \sum_{u \in U} \log p_u(x_u | a(u)).$$

The background explanation $(\emptyset, a \equiv \perp)$ has $\log \Pr = \sum_{E \in \mathcal{E}} \log(1 - \pi_E) + \sum_u \log p_u^0(x_u)$. The difference is $\sum_u g_u(a(u)) - \sum_{E \in H} w_E$. For fixed H the units do not constrain each other, so the maximum over a is taken unit by unit over the choices that (M2) admits, which gives $v_u(H)$ for unit u . The largest log posterior ratio of (H, a) over a is therefore $F_{\text{ex}}(H)$, and the maximizing assignment is $a(u) = \arg \max g_u(E)$ over $E \in H \cup \{\perp\}$ with $u \in R(E)$, where $\perp$ is excluded for $u \in R^*(H)$.

Step 2, the lower bound and its equality condition. Fix u and H and write $m_u = m_u(H)$. If $u \notin R^*(H)$, both objectives contribute $\max(0, m_u)$ and F has no penalty on u . If $u \in R^*(H)$, F_{ex} contributes m_u and F contributes $\max(0, m_u) - \sum_{E \in H: u \in R^*(E)} [-g_u(E)]_+$. In the first case $m_u \geq 0$, so the contribution of F is m_u minus a nonnegative penalty, which is equal to m_u exactly when every exact owner of u in H has $g_u(E) \geq 0$. In the second case $m_u < 0$, so every selected event that covers u has a negative ratio, F contributes $-\sum_{E \in H: u \in R^*(E)} |g_u(E)|$ and F_{ex} contributes $-\min_{E \in H: u \in R(E)} |g_u(E)|$. The former is at most the latter, with equality exactly when one selected event covers u , and that event is then its exact owner. Summing over units gives $F \leq F_{\text{ex}}$. The inequality is strict exactly when some unit of an exact representation has an owner in H with a negative ratio and is covered by at least two selected events, which is the condition in the statement.

Step 3, submodularity. Put $a_{uE} = [g_u(E)]_+ \mathbf{1}[u \in R(E)]$, which is nonnegative. The coverage term of F equals $\sum_u \max_{E \in H} a_{uE}$ with the empty maximum equal to 0. For each u , the map $H \mapsto \max_{E \in H} a_{uE}$ is nonnegative and monotone, and for $A \subseteq B$ and $E \notin B$,

$$\max_{A \cup \{E\}} a_{u\cdot} - \max_A a_{u\cdot} = [a_{uE} - \max_A a_{u\cdot}]_+ \geq [a_{uE} - \max_B a_{u\cdot}]_+ = \max_{B \cup \{E\}} a_{u\cdot} - \max_B a_{u\cdot},$$

so its marginal gains decrease. A sum of nonnegative monotone submodular functions is nonnegative, monotone and submodular. The penalty $\sum_{E \in H} \hat{w}_E$ is modular because $\hat{w}_E$ depends only on E and ϕ .

Step 4, facility location. Define $c(u, E) = G_u - [g_u(E)]_+ \geq 0$ for $u \in R(E)$ and $c(u, \perp) = G_u$, with opening cost $\hat{w}_E \geq 0$ for events and 0 for the background. For fixed H ,

$$\min_a \sum_u c(u, a(u)) = \sum_u \left(G_u - \max_{E \in H \cup \{\perp\}, u \in R(E)} [g_u(E)]_+ \right) = \sum_u G_u - \sum_u \max(0, m_u(H)),$$

and adding $\sum_{E \in H} \hat{w}_E$ gives $\sum_u G_u - F(H)$. Nonnegativity of c follows from the definition of G_u , and nonnegativity of $\hat{w}_E$ from $\pi_E \leq \frac{1}{2}$. The costs need not satisfy the triangle inequality, so the problem is a non-metric uncapacitated facility location problem with a background facility of zero opening cost.

Step 5, hardness. We reduce set cover to the maximization of F . Let a set cover instance have elements U , subsets $S_1, \dots, S_m$ and the question whether at most k subsets cover U . Create events E_i with $R^+(E_i) = S_i$ and $R^*(E_i) = \emptyset$, put $g_u(E_i) = G$ for $u \in S_i$ with a constant $G > m|U|$, which is realized by two Gaussian densities of equal variance whose means differ by $\sqrt{2G}$, evaluated at $x_u = \sqrt{2G}$, and choose $\pi_{E_i} = \frac{1}{1+e}$ so that $w_{E_i} = 1$. Then $F(H) = G|\bigcup_{E_i \in H} S_i| - |H|$ and the cost equals G times the number of uncovered elements plus $|H|$. A solution that leaves an element uncovered costs at least $G > m$, whereas the full family costs m , so the minimum cost equals the minimum cover size and the decision problems coincide. If a polynomial algorithm approximated the cost within a factor $\rho \leq (1 - o(1)) \ln|U| < |U|$, its output would cost at most $\rho m < G$, hence cover all elements, and its number of subsets would equal its cost. This would approximate set cover within ρ , which is impossible unless $P = NP$ [1, 2].

Step 6, binary responses. Let $x_u \in \{0, 1\}$, let the background rate $b_u = \Pr(x_u = 1 \mid \perp)$ tend to 0, and let the causal strengths $c_{Eu} = \Pr(x_u = 1 \mid E)$ lie in $\{0, 1\}$. A present unit has $g_u(E) = \log \frac{c_{Eu}}{b_u}$ and an absent unit $g_u(E) = \log \frac{1-c_{Eu}}{1-b_u}$. The relative likelihood of the parsimonious covering model of Peng and Reggia [3] for manifestations M^+ is

$$L(H, M^+) = \prod_{u \in M^+} \left(1 - \prod_{E \in H} (1 - c_{Eu})\right) \prod_{u \notin M^+} \prod_{E \in H} (1 - c_{Eu}) \prod_{E \in H} \frac{\pi_E}{1 - \pi_E}.$$

For $c_{Eu} \in \{0, 1\}$, $1 - \prod_{E \in H} (1 - c_{Eu}) = \max_{E \in H} c_{Eu}$, so the noisy-OR combination agrees with the dominant-cause model. As $b_u \rightarrow 0$, a present unit that no selected event covers gives $g = \log \frac{0}{b_u} \rightarrow -\infty$, in agreement with $L = 0$, so the feasible sets are exactly the covers of M^+ . An absent unit covered by a selected event with $c_{Eu} = 1$ gives $-\infty$ in both models, which forbids missing exact representations. All other factors agree term by term, so $\exp F_{\text{ex}}(H)$ equals $L(H, M^+)$ and the feasible solutions are the irredundant covers. $\square$

Corollary 1 (Forced spurious events). *If the key of every violation in $\mathcal{V}_0 \subseteq \mathcal{V}(X)$ lies only in representations of inactive events, every explanation that assigns all of $\mathcal{V}_0$ to events contains at least one inactive event, and at least the minimum hitting-set number of the sets of events whose representations contain these keys, for any nonnegative costs.*

Proof. Put $N_v = \{E : \text{key}(v) \in R(E)\}$. An explanation that assigns every $v \in \mathcal{V}_0$ to an event has $a(v) \in H \cap N_v$, so H meets every N_v and $|H \cap \bigcup_{v \in \mathcal{V}_0} N_v|$ is at least the minimum hitting-set number of $\{N_v\}_{v \in \mathcal{V}_0}$, which is at least one when $\mathcal{V}_0 \neq \emptyset$. Each N_v contains only inactive events, so all these events are spurious. Costs select among feasible explanations and do not change the feasible set, so the bound holds for every nonnegative cost. $\square$

Remark S1 (Approximation of the cost). The density greedy rule approximates the facility location cost of Theorem 1 within $H(\Delta) = \sum_{i=1}^{\Delta} \frac{1}{i}$ with $\Delta = \max(1, \max_E |R(E)|)$, and a primal-dual rule approximates it within $f = \max_u |\{E : u \in R(E)\}|$. For the greedy rule, every solution splits into stars (E, T) with $T \subseteq R(E)$ and cost $\hat{w}_E + \sum_{u \in T} c(u, E)$, and background stars $(\perp, \{u\})$ with cost G_u , so the weighted set cover over stars has an optimum no larger than the facility location optimum, and merging the stars of one facility does not increase the cost. Chvátal's greedy rule [4] then gives $H(\Delta)$, and for fixed E the star with the smallest cost per newly covered unit is a prefix of the units sorted by $c(u, E)$ [5], so each greedy step runs in polynomial time. For the primal-dual rule, raise the dual variables α_u of all unfrozen units at the same rate, raise $\beta_{uE} = \alpha_u - c(u, E)$ once α_u reaches $c(u, E)$, open E when $\sum_u \beta_{uE} = \hat{w}_E$, and freeze the units assigned to an open facility. The background is open from the start, so every unit freezes with $\alpha_u \leq G_u$. The cost equals $\sum_u \alpha_u + \sum_u \sum_{E \in O, E \neq \sigma(u)} \beta_{uE} \leq f \sum_u \alpha_u$, where O is the set of opened events and $\sigma(u)$ the facility of u , because $\beta_{uE} \leq \alpha_u$ and $\beta_{uE} > 0$ only for the at most f events that cover u , and weak duality bounds $\sum_u \alpha_u$ by the optimum. Neither factor transfers to a ratio on F , since F and the cost differ by the constant $\sum_u G_u$, and a cost within ρ of the optimum C^* only gives $F(H) \geq F^* - (\rho - 1)C^*$.

3 Theorem 2 and Corollaries 2 and 3

Theorem 2 (Interface lower bound on paired runs). *Let P and Q be window laws, T a measurable interface, and f any rule, possibly randomized, that depends on the window only through $T(X)$. For every decision event $\mathcal{A}$, $P(f \notin \mathcal{A}) + Q(f \in \mathcal{A}) \geq 1 - \text{TV}(T_{\#}P, T_{\#}Q)$, where TV is the total variation distance and $T_{\#}P$ the law of $T(X)$ under P , and for every coupling (X, X') of P and Q , $\text{TV}(T_{\#}P, T_{\#}Q) \leq \Pr(T(X) \neq T(X'))$. Paired runs of events h and h' are such a coupling, so the two errors of any rule that depends on the window only through T sum to at least $1 - \delta_T(h, h')$, where $\delta_T(h, h') = \Pr_{\xi}(T(X^{(h)}) \neq T(X^{(h')}))$ over the shared noise ξ is measured on paired runs without any distributional assumption.*

Proof. Two-point bound [6]. Put $\mu = T_{\#}P$ and $\nu = T_{\#}Q$, and let the randomization W of f be independent of the window with law λ . The event $\{f \in \mathcal{A}\}$ equals $\{(T(X), W) \in \mathcal{A}\}$ for a measurable set $\mathcal{A}$. For every w , the section A_w satisfies $|\mu(A_w) - \nu(A_w)| \leq \text{TV}(\mu, \nu)$, so

$$Q(f \in \mathcal{A}) - P(f \in \mathcal{A}) = \int [\nu(A_w) - \mu(A_w)] \lambda(dw) \geq -\text{TV}(\mu, \nu).$$

Rearranging gives $P(f \notin \mathcal{A}) + Q(f \in \mathcal{A}) = 1 - P(f \in \mathcal{A}) + Q(f \in \mathcal{A}) \geq 1 - \text{TV}(\mu, \nu)$.

Data processing. For every measurable B , $\mu(B) - \nu(B) = P(T^{-1}B) - Q(T^{-1}B) \leq \text{TV}(P, Q)$, so $\text{TV}(\mu, \nu) \leq \text{TV}(P, Q)$.

Coupling. Let (X, X') have marginals P and Q . For every measurable B ,

$$\mu(B) - \nu(B) = \Pr(T(X) \in B) - \Pr(T(X') \in B) \leq \Pr(T(X) \in B, T(X') \notin B) \leq \Pr(T(X) \neq T(X')),$$

and the supremum over B gives the claim. Paired runs produce $X^{(h)}(\xi)$ and $X^{(h')}(\xi)$ from the same noise stream ξ , and their marginals are the window laws under h and h' , so they form a coupling, which gives the bound with $\delta_T(h, h')$. Under equal priors the smallest error probability is $\frac{1}{2} \min_B [\mu(B^c) + \nu(B)] = \frac{1}{2}(1 - \text{TV}(\mu, \nu))$ by the variational form of the total variation distance. $\square$

Corollary 2 (Interval knowledge). *If two events have the same interval representation on every unit and each places its descriptions inside the box of these intervals with probability at least $1 - \epsilon$, every rule that depends only on interval membership errs with probability at least $\frac{1}{2}(1 - \epsilon)$ under equal priors, whereas the Bayes error with continuous degrees is $\frac{1}{2}(1 - \text{TV}(P_{E_1}, P_{E_2}))$.*

Proof. Let I_u be the common interval of unit u and $B = \prod_u I_u$ the box. The interval interface $T_I(\phi) = (\mathbf{1}[\phi_u \in I_u])_u$, and every interface that is constant on B , takes one value t_0 on B . The laws $\mu_i = T_{I\#}P_{E_i}$ put mass at least $1 - \epsilon$ on t_0 . For a set A that contains t_0 , both $\mu_1(A)$ and $\mu_2(A)$ lie in $[1 - \epsilon, 1]$, and for a set that does not contain t_0 , both lie in $[0, \epsilon]$. Hence $\text{TV}(\mu_1, \mu_2) \leq \epsilon$, and Theorem 2 with equal priors gives an error of at least $\frac{1}{2}(1 - \epsilon)$. The Bayes error with continuous degrees is the variational form of the total variation distance. If the in-box probability is given per unit for m units, the union bound replaces ϵ by $m\epsilon$. $\square$

Remark S2 (An example with identical intervals). Let both events place the degree on $\{4, 8\}$, with probabilities $(0.9, 0.1)$ and $(0.1, 0.9)$. The 5% and 95% quantile intervals of both are $[4, 8]$, so $\epsilon = 0$ and every rule that depends only on interval membership errs with probability $\frac{1}{2}$, whereas the continuous degree gives $\text{TV} = 0.8$ and Bayes error 0.1.

Corollary 3 (Sub-threshold events). *Every window-level rule that depends only on the violation set satisfies $\Pr_E(E \notin f) + \Pr_{\emptyset}(E \in f) \geq 1 - \delta_V(E, \emptyset)$, where $\delta_V(E, \emptyset)$ is the probability that the paired windows of E and of the fault-free run have different violation sets.*

Proof. Apply Theorem 2 with $T = \mathcal{V}(\cdot)$, $h = E$, $h' = \emptyset$ and the decision event $\{E \in f\}$. The paired violation sets differ exactly when some key is violated in one window and not in the other, that is, when E causes a violation or removes a background violation, and the union bound over these two cases bounds $\delta_V(E, \emptyset)$. $\square$

Remark S3 (Several windows of one run). If a rule uses n windows that are independent and identically distributed under each hypothesis, Le Cam's inequality $1 - \text{TV}(P, Q) \geq \frac{1}{2} A(P, Q)^2$ with the Hellinger affinity $A(P, Q) = \int \sqrt{pq}$ and the product rule $A(P^{\otimes n}, Q^{\otimes n}) = A(P, Q)^n$ give a lower bound of $\frac{1}{2} A^{2n}$ on the sum of the two errors, which decreases with n . Consecutive windows of a persistent fault are positively correlated, so the gain of accumulation over a run is measured, not derived from this bound.

4 Theorem 3 and Proposition 1

Fix a window $X = X^{(H)}(\xi)$ with true set H and $k = |H| \leq K_{\max}$. For $S \subseteq H$, the reference residual is $\psi_S = \phi(X^{(H \setminus S)}(\xi))$, the description of the paired run in which only the unselected events act, so $\psi_\emptyset = \phi(X)$ and $\psi_H = \phi(X^{(0)}(\xi))$. This definition keeps the interaction among the remaining events inside the reference, so the perturbation below contains only the error of the estimated footprints and the change of interaction caused by removing S . Along the path of Section 1, the perturbation at step $j \geq 1$ is

$$\omega_j = \max_{E \in \mathcal{E} \setminus S_j} |z_E(r_j) - z_E(\psi_{S_j})|,$$

which the algorithm and the paired runs determine without any Lipschitz constant, and ω_{S_j} denotes ω_j as in the main text. The margins are

$$\eta_\emptyset = \max_{E \in H} p_E(X) - \max_{E' \notin H} p_{E'}(X), \quad \eta_S = \max_{E \in H \setminus S} \min \left\{ z_E(\psi_S) - \max_{E' \notin H} z_{E'}(\psi_S), z_E(\psi_S) - \tau_1 \right\} \quad (\emptyset \neq S \subsetneq H),$$

and $\eta_H = \tau_1 - \max_{E \notin H} z_E(\psi_H)$.

Theorem 3 (Recovery certificate). *If $p_\emptyset(X) < \tau_0$, $\eta_\emptyset > 0$, $\eta_S > 2\omega_S$ for every prefix $\emptyset \neq S \subsetneq H$ that AEC-X visits, and $\eta_H > \omega_H$ when it reaches H , then AEC-X returns exactly H , and every violation on a unit that lies in the support of exactly one event $E \in H$ is assigned to E .*

Proof. We show by induction on j that $S_j \subseteq H$ and that the algorithm continues for $j < k$, and that it stops with $S_k = H$.

First step. Since $p_\emptyset(X) < \tau_0$, the window is not explained by the background alone. Let $E^\dagger \in H$ attain $\max_{E \in H} p_E(X)$. Since $\eta_\emptyset > 0$, $p_{E^\dagger}(X) > p_{E'}(X)$ for every $E' \notin H$, so $E_1 = \arg \max_{E \in \mathcal{E}} p_E(X)$ belongs to H and $S_1 \subseteq H$. The first step uses the window itself, so no perturbation enters it.

Step j with $1 \leq j < k$. Let $S_j \subsetneq H$ and let $E^\dagger \in H \setminus S_j$ attain η_{S_j} . By the definition of ω_j , $z_{E^\dagger}(r_j) \geq z_{E^\dagger}(\psi_{S_j}) - \omega_j$ and $z_{E'}(r_j) \leq z_{E'}(\psi_{S_j}) + \omega_j$ for every $E' \notin H$. Hence

$$z_{E^\dagger}(r_j) - z_{E'}(r_j) \geq \eta_{S_j} - 2\omega_j > 0, \quad z_{E^\dagger}(r_j) \geq \tau_1 + \eta_{S_j} - \omega_j > \tau_1.$$

The candidate $E_{j+1} = \arg \max_{E \notin S_j} z_E(r_j)$ has a score of at least $z_{E^\dagger}(r_j)$, which exceeds the score of every event outside H , so $E_{j+1} \in H \setminus S_j$, and its score exceeds τ_1 , so it is added. Since $|S_j| = j < k \leq K_{\max}$, the bound does not stop the algorithm, and $S_{j+1} \subseteq H$.

Termination. After k additions of distinct true events, $S_k = H$. For every $E \notin H$, $z_E(r_k) \leq z_E(\psi_H) + \omega_k \leq \tau_1 - \eta_H + \omega_k < \tau_1$. The candidate at this step lies outside H , so its score is below τ_1 and the algorithm stops with output H . If $k = K_{\max}$, the bound stops it with the same output.

Assignment. With output H , a violation on a unit u that lies in the support of exactly one $E \in H$ has the single candidate E , so the assignment step sends it to E . $\square$

Remark S4 (Conditions before running the algorithm). The statement is an implication along the path that AEC-X actually visits. If $\bar{\omega}_S$ is the supremum of the perturbation over residuals obtained by deducting the events of S in any order, then $\eta_\emptyset > 0$, $\eta_S > 2\bar{\omega}_S$ for every $\emptyset \neq S \subsetneq H$ and $\eta_H > \bar{\omega}_H$ imply the same conclusion, because every visited residual is obtained in one of these orders. Checking the condition on the prefixes of a single order does not suffice. Take $H = \{A, B\}$ with a false event C , and let the scores on $\psi_{\{A\}}$ favor B while the scores on $\psi_{\{B\}}$ favor C . If the first selection is A , the algorithm recovers H , and if a small change of the window makes B the first selection, it outputs $\{B, C\}$. The perturbation cannot be replaced by a Lipschitz bound for tree-based scores, since a score such as $10 \cdot \mathbf{1}[\phi_1 > 0.5]$ changes by 10 between $\phi_1 = 0.5001$ and $\phi_1 = 0.4999$.

Proposition 1 (Interaction bound and single-label rules). (i) If a learner observes only fault-free runs, single-event runs and interaction-free superpositions, then for any pair $\{A, B\}$ and third event C there are two processes that agree on all such data, one of which makes the window of $\{A, B\}$ distributed as that of C with both A and B effective, and every rule f satisfies $\Pr(f \neq \{A, B\} \mid A, B) + \Pr(f \neq \{C\} \mid C) \geq 1$ under that process. (ii) A rule that outputs one event per window has event-set F1 at most $\frac{2}{k+1}$ on windows with k effective events.

Proof. (i) Take any process $\mathcal{P}$ with single-event effects d_A, d_B, d_C and interaction-free superpositions. Define $\mathcal{P}'$ to agree with $\mathcal{P}$ on fault-free runs, single-event runs and every interaction-free superposition, and to set the run of $\{A, B\}$ to $X^{(AB)}(\xi) = X^{(0)}(\xi) + d_C(\xi)$, which corresponds to the interaction $I_{AB} = d_C - d_A - d_B$. The learner observes the same distributions under both processes, so it produces the same rule f , possibly randomized. Under $\mathcal{P}'$ the windows of $\{A, B\}$ and of C have the same law μ , so $p = \Pr_\mu(f = \{A, B\})$ is the same on both. Then $\Pr(f \neq \{A, B\} \mid A, B) = 1 - p$ and $\Pr(f \neq \{C\} \mid C) \geq \Pr(f = \{A, B\} \mid C) = p$, and the sum is at least one. To make A and B effective, choose C with a support disjoint from those of A and B and with $\mathcal{V}(X^{(0)} + d_C) \setminus \mathcal{V}(X^{(0)}) \neq \emptyset$. Then $\mathcal{V}(X^{(AB)}) \setminus \mathcal{V}(X^{(B)}) \supseteq \mathcal{V}(X^{(0)} + d_C) \setminus \mathcal{V}(X^{(0)} + d_B)$ is nonempty, so A is a cause of a violation, and B likewise. Under $\mathcal{P}'$ the interaction ratio is $\rho_{AB} = \frac{\|d_C - d_A - d_B\|}{\|d_A + d_B\|}$, so a recovery guarantee from single-event knowledge must exclude interactions of this size.

(ii) A rule that outputs $\{\ell\}$ on a window with true set T , $|T| = k$, has event-set F1 $\frac{2|\{\ell\} \cap T|}{1+k}$, which is $\frac{2}{k+1}$ if $\ell \in T$ and 0 otherwise. $\square$

5 Theorem 4

Lemma S1 (Exchangeable ranks). *Let $Z_1, \dots, Z_{n+1}$ be exchangeable, $m = \lceil (1 - \alpha)(n + 1) \rceil \leq n$, and $Z_{(m)}$ the m -th smallest of $Z_1, \dots, Z_n$. Then $\Pr(Z_{n+1} > Z_{(m)}) \leq 1 - \frac{m}{n+1} \leq \alpha$. If ties have probability zero, $\Pr(Z_{n+1} \geq Z_{(m)}) = \Pr(Z_{n+1} > Z_{(m)}) = 1 - \frac{m}{n+1}$, which lies in $(\alpha - \frac{1}{n+1}, \alpha]$.*

Proof. Put $N_i = |\{j \leq n + 1 : Z_j < Z_i\}|$. The event $Z_{n+1} > Z_{(m)}$ implies that Z_{n+1} exceeds at least m of the first n values, so $N_{n+1} \geq m$. Sorting the $n + 1$ values with ties broken arbitrarily, the value in position q has $N_i \leq q - 1$, so at most $n + 1 - m$ indices have $N_i \geq m$. By exchangeability, $\Pr(N_{n+1} \geq m) = \frac{1}{n+1} \mathbb{E}|\{i : N_i \geq m\}| \leq \frac{n+1-m}{n+1}$, and $m \geq (1 - \alpha)(n + 1)$ gives the bound α . Without ties, $\{Z_{n+1} \geq Z_{(m)}\} = \{Z_{n+1} > Z_{(m)}\} = \{N_{n+1} \geq m\}$ up to a null set, and exactly $n + 1 - m$ indices have $N_i \geq m$, so the probability equals $\frac{n+1-m}{n+1}$. From $m < (1 - \alpha)(n + 1) + 1$ this exceeds $\alpha - \frac{1}{n+1}$. $\square$

Lemma S2 (Selection by a fixed criterion). *Let $Z_1, \dots, Z_{n+1}$ be exchangeable, $\mathcal{S}$ a measurable set and $J = \{i : Z_i \in \mathcal{S}\}$. Conditionally on $J = J_0$, the variables $(Z_i)_{i \in J_0}$ are exchangeable. Hence Lemma S1 holds for the selected values given $n + 1 \in J$, with $|J| - 1$ in place of n .*

Proof. Let π permute the indices in J_0 and fix the others, and let h be bounded and measurable. The event $\{J = J_0\}$ is invariant under π , because π only moves coordinates inside J_0 , all of which lie in $\mathcal{S}$. Exchangeability gives $\mathbb{E}[h((Z_{\pi(i)})_{i \in J_0}) \mathbf{1}\{J = J_0\}] = \mathbb{E}[h((Z_i)_{i \in J_0}) \mathbf{1}\{J = J_0\}]$, which is the conditional exchangeability. $\square$

Theorem 4 (Finite-sample control of false naming). *Let fitting, calibration and test runs be disjoint, calibration and test runs exchangeable, each run contribute one window chosen independently of its content, and the calibration statistics have no ties. (a) For a fixed weight λ and τ_0 as in Section IV-B, AEC-X names an event on a fault-free test window with probability between $\alpha_0 - \frac{1}{n_0+1}$ and α_0 . (b) If τ_1 is computed on the single-event calibration windows whose first selection is correct, AEC-X adds a false event to a single-event test window whose first selection is correct with probability at most α_1 . (c) With τ_u as in Section IV-C, a test window drawn from the distribution of the calibration windows of known events is flagged unknown with probability at most α_u .*

Proof. The knowledge, the footprint operators, the scorer, the encoder, the response detectors and the addition decision are fitted on fitting runs only, and the weight λ of the first selection is fixed, so every statistic below is a fixed function of a window and its run label. Because the runs are exchangeable and each contributes one window chosen without reading its content, the calibration statistics and the test statistic of the same kind are exchangeable.

(a) Put $Z_i = 1 - p_\emptyset(X_i)$ for the n_0 fault-free calibration windows and Z_{n_0+1} for the fault-free test window. The construction of τ_0 makes $1 - \tau_0$ equal to $Z_{(m_0)}$ with $m_0 = \lceil (1 - \alpha_0)(n_0 + 1) \rceil$. AEC-X names an event exactly when $p_\emptyset(X) < \tau_0$, that is, when $Z_{n_0+1} > Z_{(m_0)}$, and Lemma S1 gives both bounds.

(b) For a single-event window with true event E_i , put $T_i^* = \max_{E' \neq E_i} z_{E'}(X_i, \phi_i - \hat{\Delta}_{E_i}(\phi_i), \{E_i\})$. On a single-event test window whose first selection is correct, the candidate at the second step is the false event with the largest addition score after the true event is deducted, so the second step adds a false event exactly when $T_{\text{test}}^* \geq \tau_1$. If the second step adds nothing, the algorithm stops with the correct output, so no later step can add a false event and no union over the steps is needed. The criterion that the first selection is correct is a fixed function of a window and its label, so Lemma S2 makes the selected calibration values and the selected test value exchangeable, and Lemma S1 gives $\Pr(T_{\text{test}}^* \geq \tau_1 \mid \text{first selection correct}) \leq \alpha_1$, where the absence of ties makes $\geq$ and $>$ equivalent.

(c) The output set $H(X)$ of AEC-X is a fixed function of the window, so the deviation $D_{H(X)}(\phi(X))$ is a fixed function as well. The calibration windows of normal operation and known single events and a test window drawn from the same distribution give exchangeable deviations, and τ_u is their conformal quantile, so Lemma S1 bounds the flag probability by α_u . $\square$

Remark S5 (Scope and fluctuation). The guarantee in (c) concerns a test window drawn from the mixture of normal and single-event windows that the calibration pool represents. A per-output threshold would give the bound conditionally on each output set, and a global threshold can exceed α_u on an output set that the calibration pool underrepresents. Windows of known concurrent events reach the template only after two estimated deductions and are not exchangeable with any calibration window, so their false-flag rate is measured in the experiments. Conditionally on the calibration set and without ties, the naming probability of (a) follows the law $\text{Beta}(n_0 + 1 - m_0, m_0)$, because $F(Z_{(m_0)})$ is the m_0 -th order statistic of n_0 uniform variables for the continuous distribution function F of the statistic [7, 8].

6 Functional Constraints

Proposition S1 (Functional constraints). *Let $x_j = \beta_0 + \beta^\top x_P + e$ hold in normal operation with coefficient of determination R^2 , and let the range and functional intervals have half-widths proportional to the normal standard deviations of x_j and e . An event that shifts x_j while x_P is unchanged produces a functional degree $\frac{1}{\sqrt{1-R^2}}$ times its range degree. An event that shifts a single partner x_i while control holds x_j produces $\frac{|R|}{\sqrt{1-R^2}}$ times the range degree of x_i , which exceeds one if and only if $R^2 > \frac{1}{2}$. An event that moves x_j along the relation leaves the functional constraint silent.*

Proof. Let e be uncorrelated with x_P , so that $\text{Var}(e) = \sigma_j^2(1 - R^2)$ with $\sigma_j^2 = \text{Var}(x_j)$, and let the range and functional half-widths be $c\sigma_j$ and $c\sigma_j\sqrt{1-R^2}$ with the same factor c . An event that shifts x_j by Δ_j and x_P by Δ_P shifts the residual by $\Delta_j - \beta^\top \Delta_P$, so the functional degree is $\frac{\Delta_j - \beta^\top \Delta_P}{c\sigma_j\sqrt{1-R^2}}$ up to the noise term, which has the same order in both constraints. If $\Delta_P = 0$, the range degree is $\frac{\Delta_j}{c\sigma_j}$ and the ratio is $\frac{1}{\sqrt{1-R^2}}$. With a single partner, $\beta = R\frac{\sigma_j}{\sigma_i}$, and if $\Delta_j = 0$ the functional degree has magnitude $\frac{|R|\sigma_j|\Delta_i|}{\sigma_i c\sigma_j\sqrt{1-R^2}} = \frac{|\Delta_i|}{c\sigma_i} \cdot \frac{|R|}{\sqrt{1-R^2}}$, and $\frac{|R|}{\sqrt{1-R^2}} > 1$ holds exactly when $R^2 > \frac{1}{2}$. If $\Delta_j = \beta^\top \Delta_P$, the residual does not move and the functional degree is zero. $\square$

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